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Quillstone Payments
Customer Operations AI Team

Application: Machine Learning Engineer - Customer Support AI

Dear Hiring Team,

At Thistlebox, I build models that inform weekly operational decisions and evaluate whether customer interventions work. I am applying for Quillstone Payments' Machine Learning Engineer - Customer Support AI role to bring that combination of Python development, feature engineering and rigorous evaluation to your team. Your work helping merchants get answers quickly connects practical operational outcomes with my intended move from data science into machine learning engineering.

My demand forecasting model predicts orders by recipe and region six weeks ahead, reducing WAPE from 23% to 14% and contributing to an estimated 9% reduction in food waste. It runs as a scheduled Vertex AI Workbench notebook and writes results to BigQuery for operations to use through Looker. I also developed a weekly batch-scored XGBoost churn model; evaluation with a 10% holdout showed churn 2.1 percentage points lower in the treated group. These projects demonstrate my experience preparing model features, building repeatable data workflows and assessing models against business outcomes.

Alongside modelling, I write production SQL, maintain dbt models and use Git. Since 2025, I have introduced pytest tests for forecasting feature code, reaching approximately 60% coverage. For a Streamlit experimentation tool used by around 25 colleagues, I wrote both the application and its Dockerfile, while the data platform team handles deployment. My delivery experience is in scheduled and batch workflows rather than independently deployed model-serving APIs. I would bring that practical foundation and a strong statistical background to collaboration with your engineers, data scientists and product managers.

Quillstone's emphasis on sharing context and learning is reflected in how I work: I present findings to leadership monthly and have delivered "Stats for PMs" training three times. I would welcome a conversation about how my modelling, evaluation and data-processing experience could contribute to your customer-support AI systems as I develop further in production ML engineering. Thank you for considering my application.

Kind regards,
Tobias Mwangi-Hartley